

RECIFE, Brazil

WMO: 828990

Lat: 8.126S

Lon: 34.924W

Elev: 10

StdP: 101.20

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	22.0	17.8	12.8	25.0	18.0	13.0	25.1	10.3	26.7	9.5	26.5	2.5	230	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.3	34.0	27.1	33.2	26.6	32.8	26.4	27.6	32.4	27.1	31.8	26.7	31.2	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	30.7	25.9	21.3	30.4	25.2	20.4	29.5	87.6	32.5	85.6	32.3	84.1	31.4	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.7	8.1	7.4		DB	20.4	33.8	1.1	2.1	19.6	35.3	18.9	36.5	18.3	37.7	17.5	39.3
				WB	18.7	27.9	1.3	1.4	17.8	28.9	17.0	29.7	16.3	30.4	15.4	31.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	28.4	28.8	28.6	28.0	27.2	26.2	25.7	25.7	26.5	27.5	28.1	28.4	(6)
	DBStd	1.64	1.34	1.39	1.40	1.31	1.21	1.15	1.13	1.14	1.18	1.17	1.07	1.09	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6356	571	526	577	539	534	486	486	487	496	543	542	570	(10)
	CDD18.3	3314	312	292	319	289	276	236	227	228	246	285	292	312	(11)
	CDH23.3	34860	3673	3572	3932	3320	2832	1992	1695	1803	2234	2959	3244	3604	(12)
	CDH26.7	11108	1353	1441	1614	1229	864	448	328	338	463	788	1002	1240	(13)
(14) Wind	WSAvg	4.3	4.1	4.1	3.8	3.6	3.7	4.0	4.5	4.8	5.0	4.7	4.5	4.3	(14)
(15) Precipitation	PrecAvg	2303	105	136	237	297	325	377	339	198	111	54	43	65	(15)
	PrecMax	3857	275	376	626	771	756	709	675	626	336	149	221	201	(16)
	PrecMin	1453	0	19	27	57	60	125	64	47	20	0	3	7	(17)
	PrecStd	492	69	99	135	166	132	151	145	103	74	40	35	50	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.4	34.9	33.8	33.1	31.7	31.1	31.1	31.9	33.1	33.6	34.1	(19)
		MCWB	27.0	27.3	27.6	27.6	26.8	26.0	25.4	25.3	25.5	26.0	26.6	26.9	(20)
	2%	DB	33.1	33.8	33.8	32.9	32.0	30.8	30.1	30.1	30.9	32.1	32.8	33.0	(21)
		MCWB	26.4	26.9	27.2	26.9	26.5	25.6	24.9	24.7	25.0	25.7	26.1	26.4	(22)
	5%	DB	32.1	32.9	32.8	31.9	31.0	29.9	29.2	29.4	29.9	31.1	31.2	31.9	(23)
		MCWB	25.9	26.4	26.7	26.4	26.1	25.0	24.5	24.3	24.4	25.2	25.2	25.7	(24)
	10%	DB	31.0	31.7	31.2	31.0	30.1	29.0	28.3	28.3	29.0	29.9	30.1	30.5	(25)
		MCWB	25.5	25.9	25.9	25.9	25.4	24.4	23.9	23.7	23.8	24.4	24.5	25.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	28.1	28.3	28.0	27.5	26.6	26.3	26.0	26.2	26.7	27.0	27.3	(27)
		MCDB	32.6	33.3	33.5	32.6	31.5	30.4	29.7	29.9	30.5	31.8	32.7	32.8	(28)
	2%	WB	26.8	27.2	27.5	27.2	26.8	26.0	25.5	25.2	25.5	26.1	26.3	26.7	(29)
		MCDB	32.0	32.3	32.3	31.4	30.7	29.6	28.9	29.0	29.6	31.4	31.8	31.9	(30)
	5%	WB	26.3	26.7	26.9	26.7	26.4	25.5	25.0	24.7	24.9	25.4	25.7	26.1	(31)
		MCDB	31.2	31.5	31.5	30.6	30.0	28.8	28.3	28.3	28.8	30.2	30.6	30.7	(32)
	10%	WB	25.9	26.2	26.5	26.4	25.9	25.0	24.5	24.2	24.3	24.8	25.1	25.5	(33)
		MCDB	30.3	30.7	30.7	30.0	29.3	28.0	27.6	27.6	28.0	29.0	29.5	29.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.3	5.7	5.8	5.8	5.6	5.7	5.7	5.0	4.8	4.9	4.9	(35)
		MCDBR	6.4	6.7	7.0	6.9	6.9	6.6	6.6	6.7	6.1	6.6	6.4	6.7	(36)
	5% WB	MCWBR	2.6	2.7	2.8	3.0	3.0	3.0	3.1	3.1	2.8	2.8	2.7	2.8	(37)
		MCWBR	5.8	6.1	6.3	6.3	6.3	6.1	6.2	6.2	5.6	5.8	5.8	5.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.394	0.398	0.397	0.393	0.386	0.390	0.391	0.395	0.425	0.421	0.399	0.404	(40)
		taud	2.503	2.497	2.507	2.538	2.559	2.534	2.517	2.471	2.361	2.377	2.451	2.455	(41)
	Ebn at Noon		946	938	925	900	879	859	866	886	885	906	934	933	(42)
		Edh at Noon	114	115	112	104	97	97	100	109	127	128	119	119	(43)
(44) All-Sky Solar Radiation	RadAvg		6.19	6.19	6.08	5.51	4.87	4.41	4.65	5.20	5.86	6.25	6.62	6.40	(44)
	RadStd		0.40	0.30	0.32	0.39	0.39	0.29	0.33	0.24	0.30	0.29	0.28	0.27	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	-0.79	-0.45	-0.79	-1.73	-0.65	-0.37	N/A	N/A	-331	-331
Regional (3 neighbors)	-0.64	-0.52	-0.93	-0.79	-0.66	-0.60	N/A	N/A	-264	-266

Nomenclature: See separate page